



06 AUTONOMOUS APPLICATION BUILDER



Problem Explanation:

Imagine a system where a user simply describes their idea—for example, "Create an e-commerce site for handmade crafts"—and a network of specialized AI agents takes over. Using the tools provided by [agent.ai](https://agent.ai/builders)(<https://agent.ai/builders>), your challenge is to develop a collaborative multi-agent system that automatically designs, builds, and deploys a web application or website based solely on the prompt.

Key Agent Roles:

- **Designer Agent:** Responsible for generating visual layouts, UI/UX design components, and branding elements.
- **Developer Agent:** Writes the backend and frontend code, selects frameworks, and ensures that all components work seamlessly together.
- **Project Manager Agent:** Oversees the project, coordinates the communication between agents, and manages deployment pipelines (e.g., setting up hosting, integrating CI/CD).

The agents must communicate and collaborate effectively to produce a final deliverable—a deployed, fully functional application that goes beyond merely generating code.

Evaluation Metrics:

A GitHub repository link is expected as your submission, containing a README file with clear instructions on how to run your code. (mandatory) Deployment of your code will earn you bonus points.

1. **End-to-End Functionality:** The final product must be a deployed application that meets the user's requirements.
2. **Design Quality & Usability:** The UI/UX should be visually appealing and intuitive.
3. **Code Quality & Integration:** Agents must generate maintainable, scalable code with minimal errors and seamless integration between components.
4. **Agent Collaboration Efficiency:** How effectively do the agents coordinate tasks, share information, and resolve conflicts?
5. **Deployment Robustness:** The solution should include automated testing, error handling, and a smooth deployment process (e.g., using CI/CD pipelines).

Examples :

- **Scenario 1:**

- Prompt: “Build a restaurant reservation web app with online ordering.”
- Outcome:
 - The Designer Agent creates an attractive, responsive UI.
 - The Developer Agent sets up the database, APIs, and front-end functionality (e.g., reservation system, menu browsing).
 - The Project Manager Agent orchestrates testing and deploys the app on a cloud service with automated scaling.

• **Scenario 2:**

- **Prompt:** “Create a personal blog with integrated social media feeds and dynamic content.”
- **Outcome:**
 - The agents produce a modern, blog-ready design; integrate content management and social media APIs; and deploy the website with continuous updates.

What We Expect:

- **Front-End:** React, Vue.js, or Angular for responsive user interfaces.
- **Back-End:** Node.js, Python (Flask, Django), or similar for API development.
- **Database:** PostgreSQL, MongoDB, or Firebase for data management.
- **Agent Framework:** agent.ai’s platform for creating and managing AI agents.
- **Deployment:** Docker, Kubernetes, or serverless platforms (e.g., AWS Lambda, Vercel) along with CI/CD pipelines (GitHub Actions, Jenkins).
- **Communication Protocols:** RESTful APIs or WebSocket for inter-agent communication.

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